

INDU 211 · Comprehensive Topic Guide (Part 8)

Chapters 6 & 11: Work Design, Work Measurement & Human Factors

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Built on the teacher's Lecture 12 (Chapters 6 and 11, Work Design and Human Factors), with depth from the course textbook (Hicks, Chapter 6: Work Design and Organizational Performance, and Chapter 11: Human Factors).

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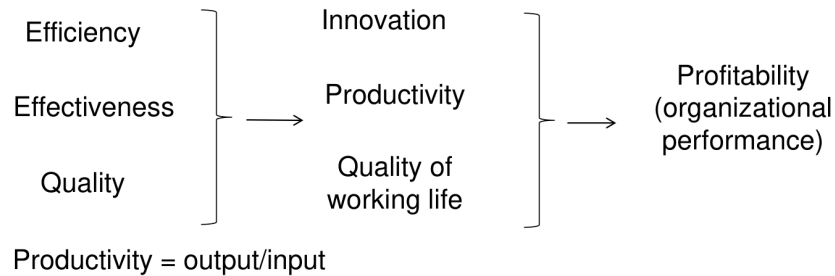
1. The Problem Work Design Solves

Suppose a plant must produce **10,000 units a year** (Lecture 12, slide 3). Before it can start, it must answer: how much equipment and how many workers? What is the most efficient work procedure? How do we train operators and pay them? How efficient are we, and what should we charge? Almost every one of these answers depends on knowing **how long the work takes**, which is why the slide ties them all to **time study**.

2. Productivity and Organisational Performance

Organization performance

- The need for an efficient and effective control system



- Work measurement: the standard by which we measure efficiency
- Work design: helps determine efficiency

Figure 1: Organization performance, Lecture 12, slide 4.

Reading the figure: efficiency, effectiveness, quality, innovation, productivity and quality of working life all feed **profitability** (organisational performance). **Work measurement** is the standard by which efficiency is measured; **work design** determines it.

$$\text{Productivity} = \frac{\text{output}}{\text{input}}$$

Measures of Productivity

Partial measures	$\frac{\text{Output}}{\text{Labor}}$	$\frac{\text{Output}}{\text{Machine}}$	$\frac{\text{Output}}{\text{Capital}}$	$\frac{\text{Output}}{\text{Energy}}$
Multifactor measures	$\frac{\text{Output}}{\text{Labor} + \text{Machine}}$		$\frac{\text{Output}}{\text{Labor} + \text{Capital} + \text{Energy}}$	
Total measure	$\frac{\text{Goods or Services Produced}}{\text{All inputs used to produce them}}$			

Figure 2: Measures of productivity, Lecture 12, slide 5.

Measure	Formula	Example
Partial	Output ÷ one input	Units per labour-hour; units per machine-hour
Multifactor	Output ÷ several inputs	Output ÷ (labour + machine)
Total	All goods and services ÷ all inputs	Whole-plant productivity

Example: 4,000 units with 800 labour-hours → labour productivity = $4000/800 = 5$ units per labour-hour. Inputs must be in one unit: you cannot add labour-hours to dollars.

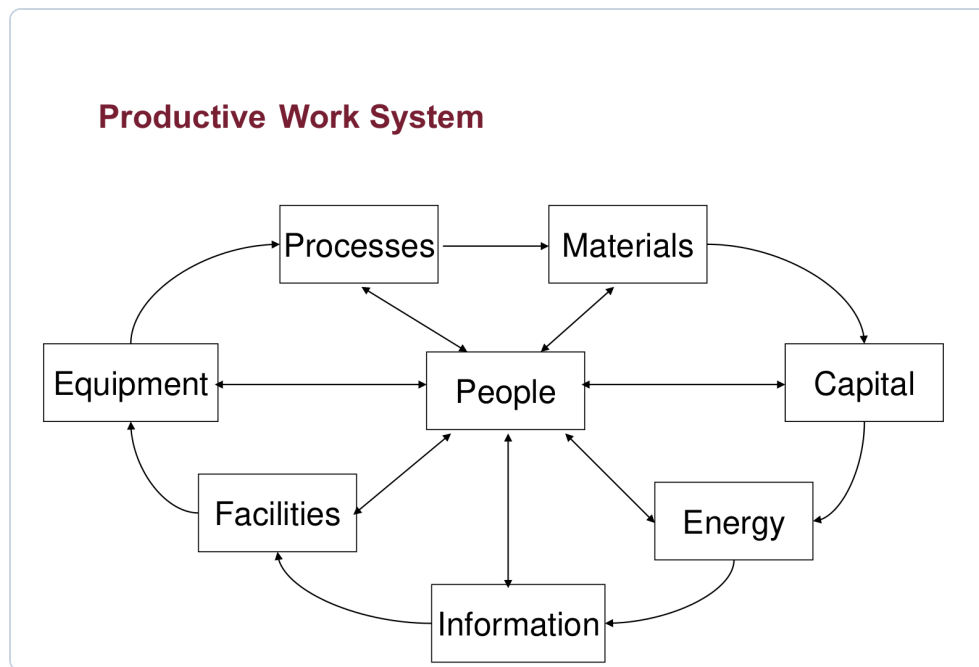


Figure 3: Productive work system, Lecture 12, slide 6.

Reading the figure: people sit at the centre, connected to processes, materials, capital, energy, information, facilities and equipment. Every component interacts with the human operator, which is why work design and human factors are studied together.

3. Human Factors (Ergonomics)

Definition (International Ergonomics Association, slide 8): ergonomics (or human factors) is the scientific discipline concerned with the understanding of interactions among humans and other elements of a system, and the profession that applies theory, principles, data and methods to design in order to optimise human well-being and overall system performance.

Human factors matches **system characteristics** to **human characteristics**, and human characteristics come from two sciences (slide 9):

Physiology (the body)	Psychology (the mind)
Skeleton, skeletal muscles, nervous system, metabolism	Motivation, boredom, stress
Lifting, reaching, carrying, hearing, seeing, speaking	"Humans like money, don't want to be bored, have brains, deserve information, fair treatment and respect, and make mistakes" (slide 10)

Machines vs humans (slides 13–14): compare them on range of operations, workpiece size, speed, tolerance capability, energy use and maintenance. Machines win on speed, force and repeatability; humans win on judgement, adaptability and handling the unexpected. Good design gives each the tasks it does best.

From the textbook (Hicks, §11.1): a productive work system combines equipment, facilities, energy, materials, information and people; the industrial engineer designs them to operate together to achieve the outputs desired, and the human component is the one that most often limits performance when it is ignored.

4. Anthropometry: Designing for Real Bodies

Anthropometry is the scientific study of the measurements and proportions of the human body (slide 15). People come in all shapes and sizes, so the first design question is: **who are you designing for?**

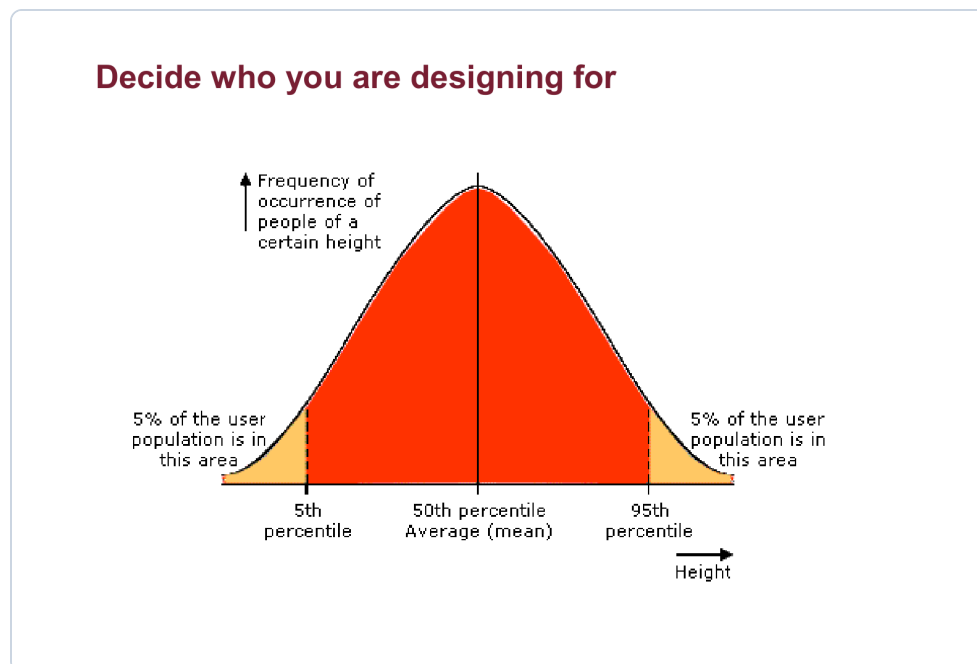


Figure 4: Decide who you are designing for, Lecture 12, slide 17.

Reading the bell curve: a body dimension (e.g. height) is roughly normal. The 50th percentile is the average; 5% of users are below the **5th percentile** and 5% above the **95th**. Designing for the 5th–95th range accommodates 90% of users.

Decide who you are designing for

What is it that you are aiming for with your design?	Design examples:	Examples of measurements to consider:	Users that your design should accommodate:
Easy reach	Vehicle dashboards, Shelving	Arm length, Shoulder height	Smallest user: 5th percentile
Adequate clearance to avoid unwanted contact or trapping	Manholes, Cinema seats	Shoulder or hip width, Thigh length	Largest user: 95th percentile
A good match between the user and the product	Seats, Cycle helmets, Pushchairs	Knee-floor height, Head circumference, Weight	Maximum range: 5th to 95th percentile
A comfortable and safe posture	Lawnmowers, Monitor positions, Worksurface heights	Elbow height, Sitting eye height, Elbow height (sitting or standing?)	Maximum range: 5th to 95th percentile
Easy operation	Screw bottle tops, Door handles, Light switches	Grip strength, Hand width, Height	Smallest or weakest user: 5th percentile
To ensure that an item can't be reached or operated	Machine guarding mesh, Distance of railings from hazard	Finger width Arm length	Smallest user: 5th percentile Largest user: 95th percentile

Figure 5: Design target by purpose, Lecture 12, slide 18.

Design goal	Examples	Design for
Easy reach	Dashboards, shelving	Smallest user: 5th percentile
Adequate clearance	Manholes, cinema seats	Largest user: 95th percentile
Good match	Seats, helmets	Adjustable: 5th–95th range
Comfortable posture	Work surfaces, monitors	Adjustable: 5th–95th range
Easy operation	Bottle tops, handles	Weakest user: 5th percentile
Prevent reaching a hazard	Guard mesh, railings	Finger width: smallest; arm length: largest

The logic: design for the user who is *limited* by the dimension. If the shortest user can reach the shelf, everyone can; if the largest fits through the manhole, everyone does. Designing for "the average person" fails half the population on the critical side.

5. Human Interfaces with the World of Work

Interface	Optimise for	Examples
Work environment	Comfort, visibility, access, safety	Desk and dial layout, illumination, temperature, humidity, air
Machines	Visibility, ingress/egress, reach and grasp, foot pedals, maintenance, strength	Cockpits, vehicle cabs
Information systems	Access, understanding, presentation	Displays, reports, buzzers; auditory vs visual signals
Organisation	Job satisfaction, morale, motivation	Job design, reporting, evaluation and reward, training

Working conditions (slides 30–31): safety and security, causes of accidents, noise and vibration, work breaks, temperature and humidity, ventilation, illumination, colour.

6. Motivation and Pay

Motivation and Money

- Taylor's scientific management (1911)
 - Workers are motivated mainly by money
 - Suggested piece-rate system (payment per unit)
- Maslow's theory (1943)
 - People are motivated by hierarchy of needs, which includes money
- Herzberg (1959)
 - Theory of motivation
 - Importance of job enrichment, job rotation, employee empowerment and recognition



Figure 6: Motivation and money, Lecture 12, slide 25.

Theorist	Year	Idea
Taylor (scientific management)	1911	Workers are motivated mainly by money → piece-rate pay
Maslow	1943	A hierarchy of needs ; money is only one level
Herzberg	1959	Job enrichment, rotation, empowerment and recognition motivate

Reading the pyramid: Maslow's lower levels (physiological, safety) are **basic needs**; the upper levels (esteem, self-actualisation) are **growth needs**. Pay satisfies the basic levels; the upper ones need meaningful work.

Monetary incentives (slide 26): bonuses (cash, stock options), profit sharing, and incentive systems: **measured daywork** (pay based on standard time) or **piece rate** (pay per piece).

7. Job Design: Specialisation vs Job Expansion

Work design specifies the tasks in a job: what, how, why and where (slide 28). Its components are working conditions, job specialisation, behavioural approaches to job design, methods analysis and work measurement.

Job specialisation (Adam Smith, 1776, the pin factory) breaks a job into small parts done by specialists.

Disadvantages	
For Management: 1. Difficult to motivate quality 2. Worker dissatisfaction, possibly resulting in absenteeism, high turnover, disruptive tactics, poor attention to quality	For Labor: 1. Monotonous work 2. Limited opportunities for advancement 3. Little control over work 4. Little opportunity for self-fulfillment

Figure 7: Specialization: disadvantages, Lecture 12, slide 34.

	For management	For labour
Advantages (slide 33)	Simpler training, high productivity, low wage cost	Low skill needed, few responsibilities, little mental effort
Disadvantages (slide 34)	Hard to motivate quality; dissatisfaction → absenteeism, turnover, poor quality	Monotony, limited advancement, little control, little self-fulfilment

Job expansion (behavioural approach, slide 35) adds variety to reduce boredom:

Method	What changes	Direction
Job enlargement	A larger portion of the total task	Horizontal (more tasks of the same kind)
Job enrichment	More responsibility for planning and coordination	Vertical (more control)
Job rotation	Workers periodically exchange jobs	Between jobs
Employee empowerment	Authority to make decisions	Vertical

Exam tip (Fall 2020 final): management should consider **both** specialisation and job expansion; neither always dominates.

8. Methods Analysis: Charts and Motion Study

Work method simplification is a systematic search for easier, better ways to do a task: "work smart, not hard". It is the essential step **before** time study (there is no point timing a bad method). The five steps (slide 38): select a job → get and record the facts → **question every detail** → develop and test a better method → install and maintain it.

Flow process chart

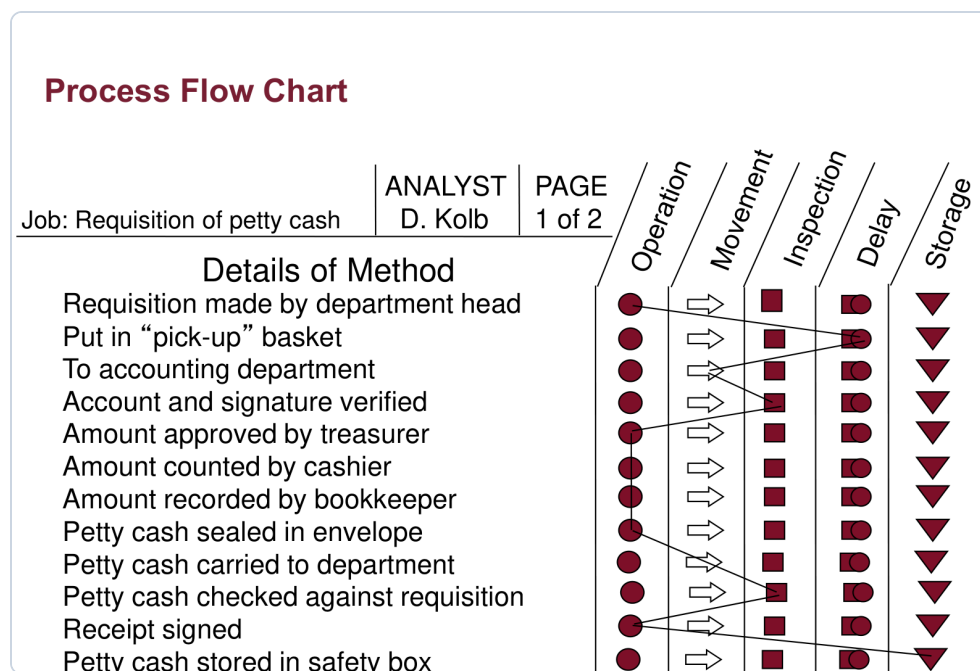


Figure 8: Process flow chart, requisition of petty cash, Lecture 12, slide 40.

Reading the chart: each row is one step of the job; the columns are the five symbols: **operation** (circle), **movement/transport** (arrow), **inspection** (square), **delay** (D) and **storage** (triangle). The analyst marks one symbol per step, and the resulting line shows at a glance how many steps are delays, transports and inspections: the non-value-added ones to attack.

For every step ask (slide 39): Is it necessary, or can it be eliminated? Can it be combined with another? Is the sequence right? Can it be improved? Is the right person doing it?

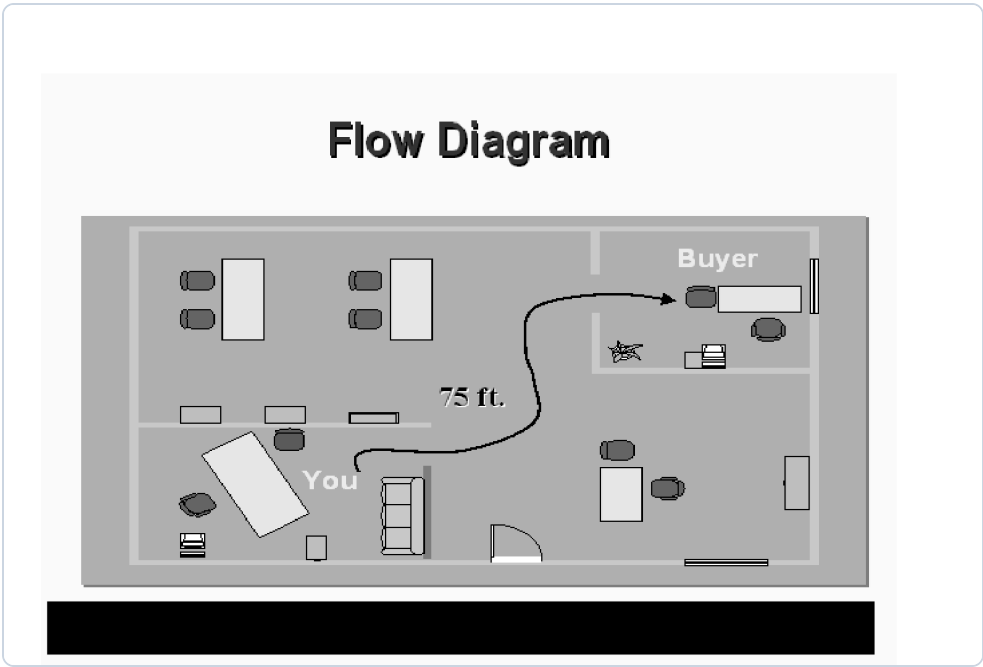


Figure 9: Flow diagram, Lecture 12, slide 42.

Reading the figure: the flow diagram draws the same steps on the **floor plan**, so long or back-tracking movements are visible as long lines across the room.

Left-hand–right-hand chart



Figure 10: Operations chart (left hand/right hand), Lecture 12, slide 44.

Reading the chart: it follows the two hands of **one operator** at **one workstation**, step by step. In the example, the right hand is idle while the left hand reaches for and grasps a bolt: a sign the work is unbalanced. It suits short, repetitive work.

Motion study (slides 47–48)

Motion study analyses human motions using **therbligs** (basic elemental motions: reach, grasp, move...) and **micro-motion study** (slow-motion film). Guidelines:

- **Eliminate** unnecessary steps and motions, using a hand as a holding device, abnormal motions, idle time.
- **Combine** motions into one continuous curved motion; combine tools and controls.
- **Rearrange** to distribute work evenly between the two hands.
- **Simplify:** reduce eye travel, use momentum, shorten motions.

From the textbook (Hicks, §6.2.4): these are the **principles of motion economy**: use both hands simultaneously and symmetrically, keep tools and materials in fixed, close positions, use gravity-feed bins and drop deliveries, and let fixtures hold the work instead of the hands.

9. Work Measurement: Setting a Standard Time

Work measurement determines the time needed to produce one unit, giving **labour standards** for costing, bids, staffing, production planning, incentive pay and efficiency (slides 49–50).

Term	Meaning
Observed (actual) time	Average of the recorded times
Performance rating (PR)	How fast the observed operator works relative to normal pace (1.13 = 13% faster than normal)
Normal time	Time for an average trained operator at a normal pace
Allowances	Time added for personal needs, fatigue and unavoidable delays
Standard time	Normal time plus allowances

$$OT = \frac{\sum x_i}{n}, \quad NT = OT \times PR, \quad ST = NT \times AF \quad (AF = 1 + \text{allowance})$$

Lecture Example 1 (slides 54–55)

Nine observations total 10.35 min; PR = 1.13; allowance 20%.

$$OT = \frac{10.35}{9} = 1.15 \text{ min}, \quad NT = 1.15 \times 1.13 = 1.30 \text{ min}, \quad ST = 1.30 \times 1.20 = 1.56 \text{ min}$$

Why multiply by PR > 1? A fast operator finishes quicker than normal, so a normal operator would need **longer**: the normal time is greater than the observed time.

Lecture Example 2 (slide 56): check the arithmetic!

Elements $15 + 7 + 5 + 8 = 35$ min; allowances 12%.

- PR = 115%: $NT = 35 \times 1.15 = 40.25$, $ST = 40.25 \times 1.12 = 45.08$ min.
- PR = 90%: $NT = 35 \times 0.90 = 31.5$ (the slide misprints this as 28.5), so $ST = 31.5 \times 1.12 = 35.28$ min (not 31.92).

From the textbook (Hicks, §6.3): besides direct time study, standards can come from **standard data** (tables of element times reused across jobs) and **predetermined time systems** (such as MTM), which assign times to basic motions, so a job can be timed before it is ever performed.

10. Exam Checklist

- [] Productivity = output/input; partial vs multifactor vs total; consistent units.
- [] Human factors rests on physiology and psychology.
- [] Percentile rule: reach → 5th, clearance → 95th, adjustable → 5th–95th.
- [] Taylor (money, piece rate), Maslow (hierarchy), Herzberg (enrichment).
- [] Specialisation pros and cons; enlargement (horizontal) vs enrichment (vertical) vs rotation.
- [] Method improvement comes before time study; the five questions on a flow process chart.
- [] $OT \rightarrow NT = OT \times PR \rightarrow ST = NT \times (1 + \text{allowance})$.